88723165

Computer Network Technology

CH10: Access Control Lists

ACL คืออะไร?

Access Control List (ACL) เป็นกลไกที่ใช้ควบคุมการเข้าถึงเครือข่ายโดยการกรองแพ็กเก็ตที่เข้าหรือออกจากเราเตอร์หรืออุปกรณ์เครือข่ายอื่น ๆ สามารถใช้ ACL เพื่อกำหนดว่าข้อมูลประเภทใดที่สามารถผ่านเครือข่ายได้

ประเภทของ ACL

1. Standard ACL (หมายเลข 1–99 และ 1300–1999)

- กรองแพ็กเก็ตตาม source IP address เท่านั้น

2. Extended ACL (หมายเลข 100–199 และ 2000–2699)

- กรองแพ็กเก็ตตาม source และ destination IP, protocol (TCP, UDP, ICMP), port number

รูปแบบการตั้งค่า ACL

1. Number ACL

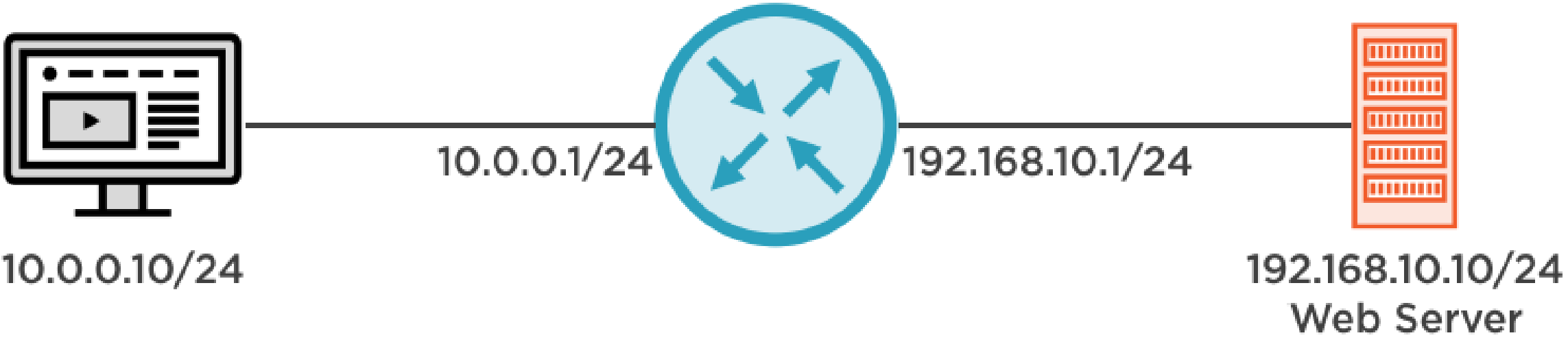
- Standard -> 1-99
- Extend -> 100-199

2. Name ACL

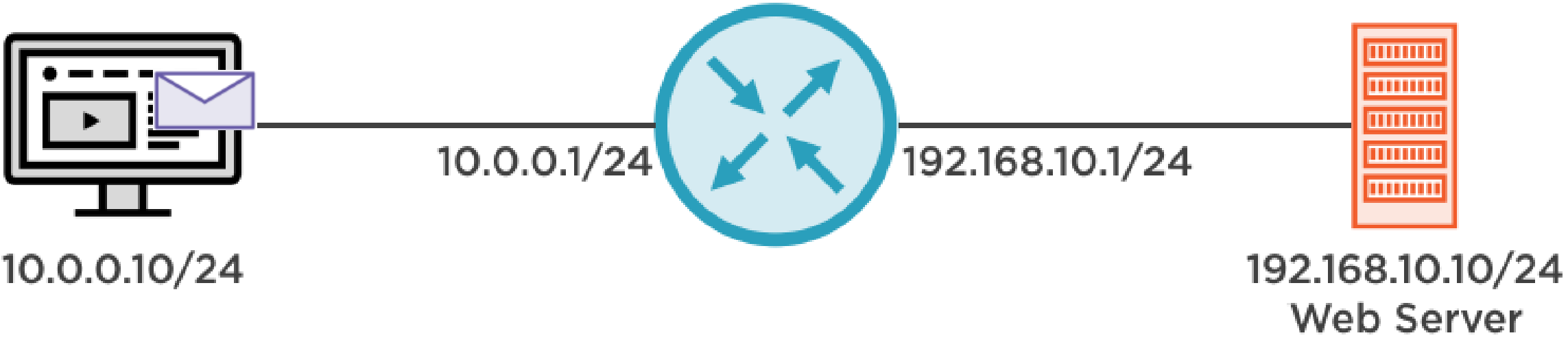
- ใช้การตั้งชื่อแทนตัวเลข

Access Control Lists Selection Criteria

Standard Access Control Lists



Standard Access Control Lists



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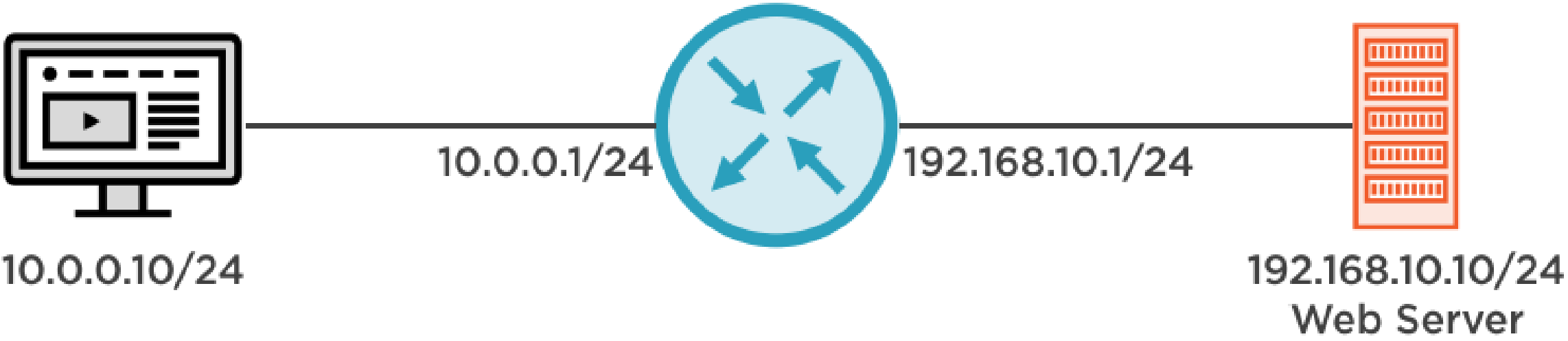
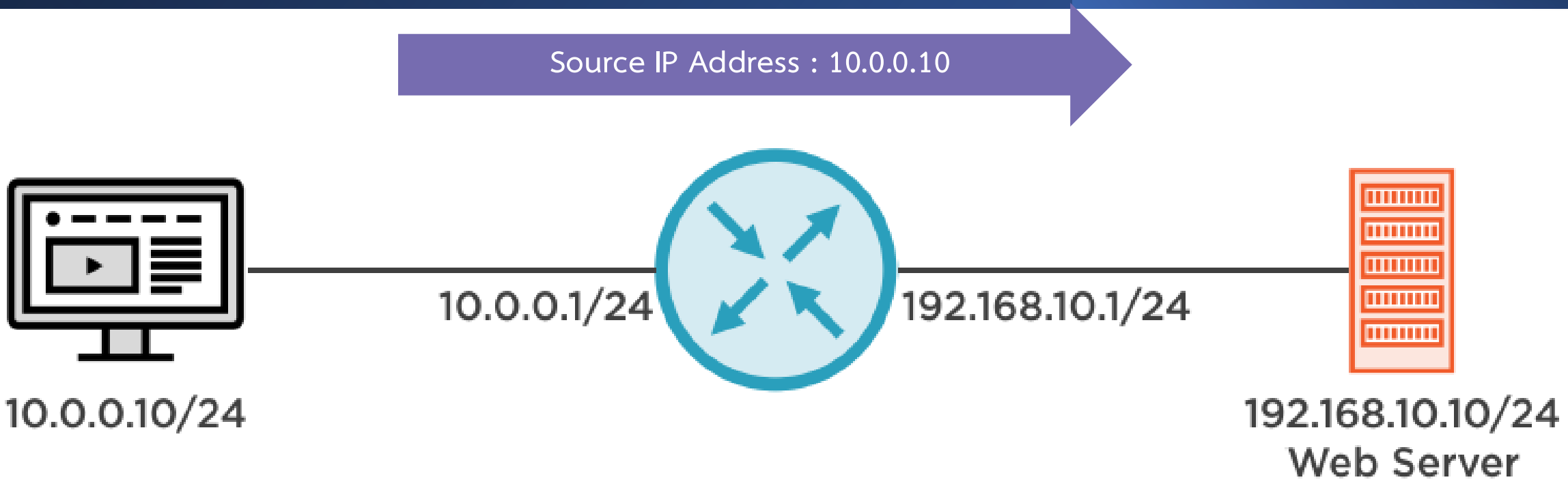


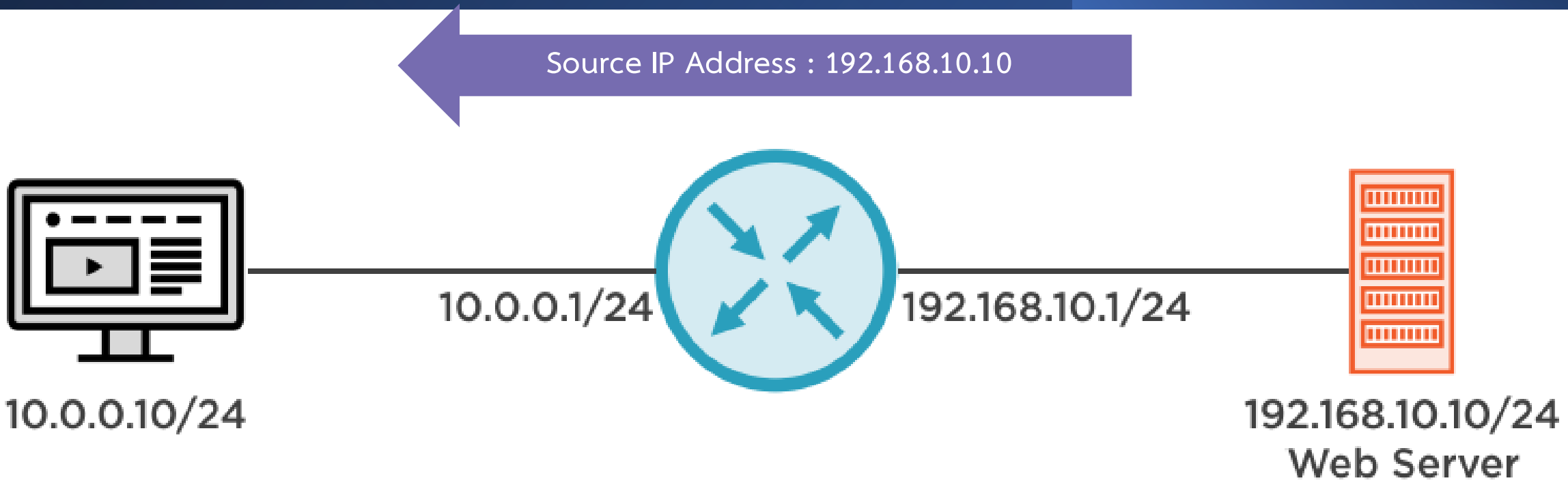
Source IP Address : 10.0.0.10

Destination IP Address : 192.168.10.10

Standard Access Control Lists



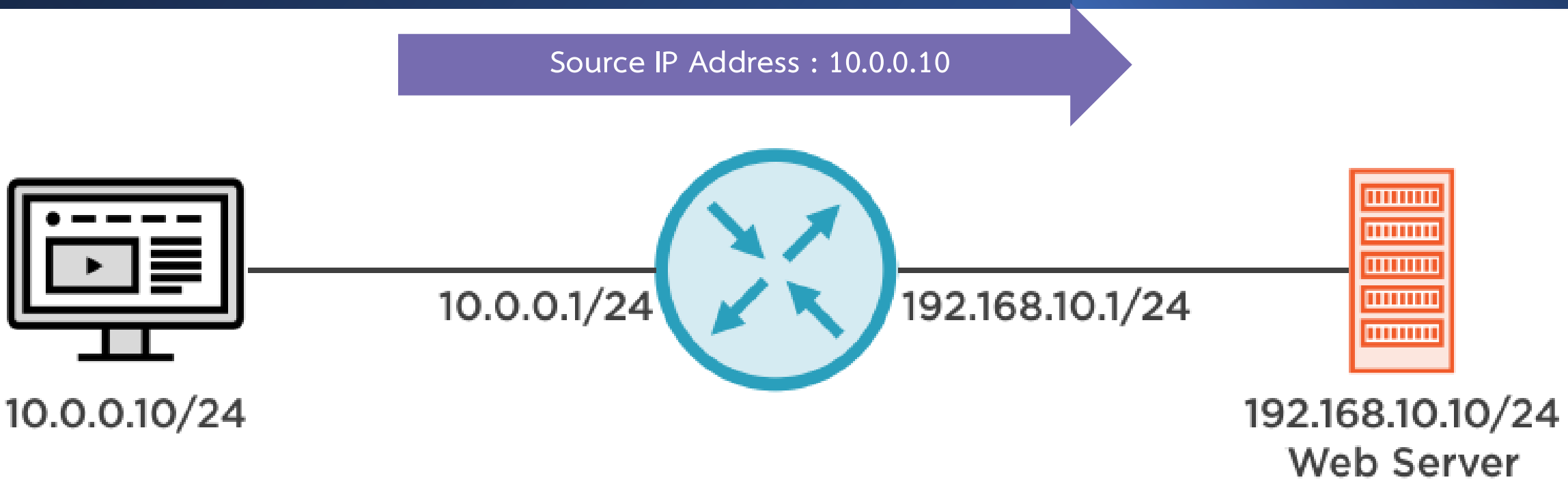
Standard Access Control Lists



ขั้นตอนการตั้งค่า ACL

1. สร้าง ACL
2. ประกาศใช้ ACL (Interface)

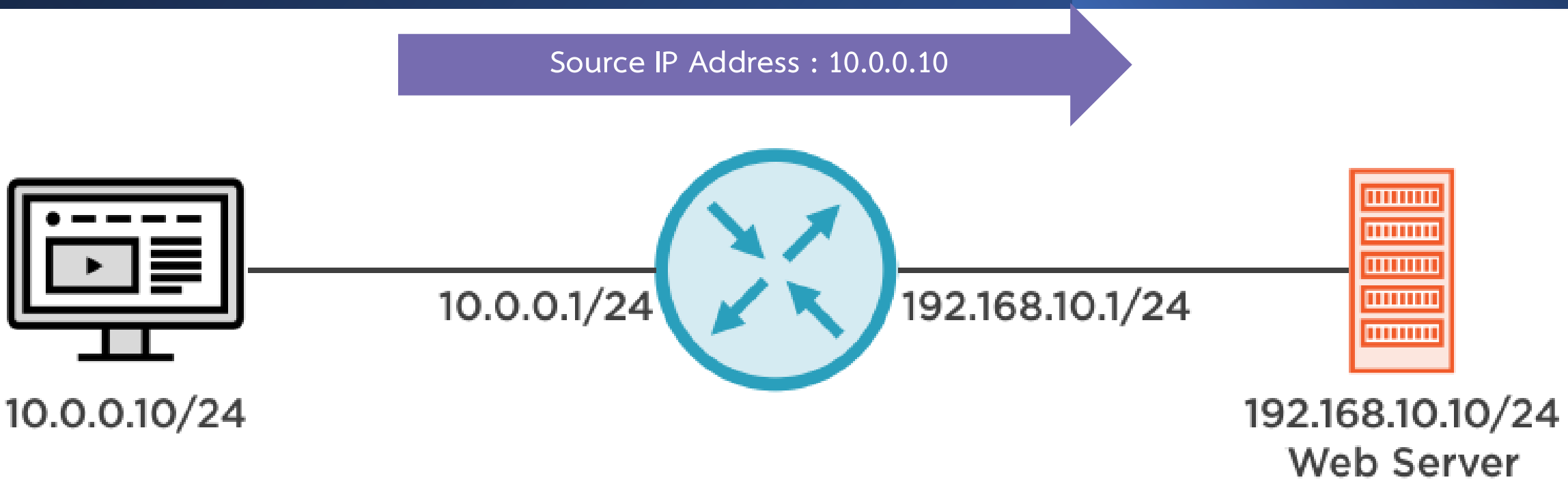
Standard Access Control Lists



1. สร้าง ACL

permit host 10.0.0.10

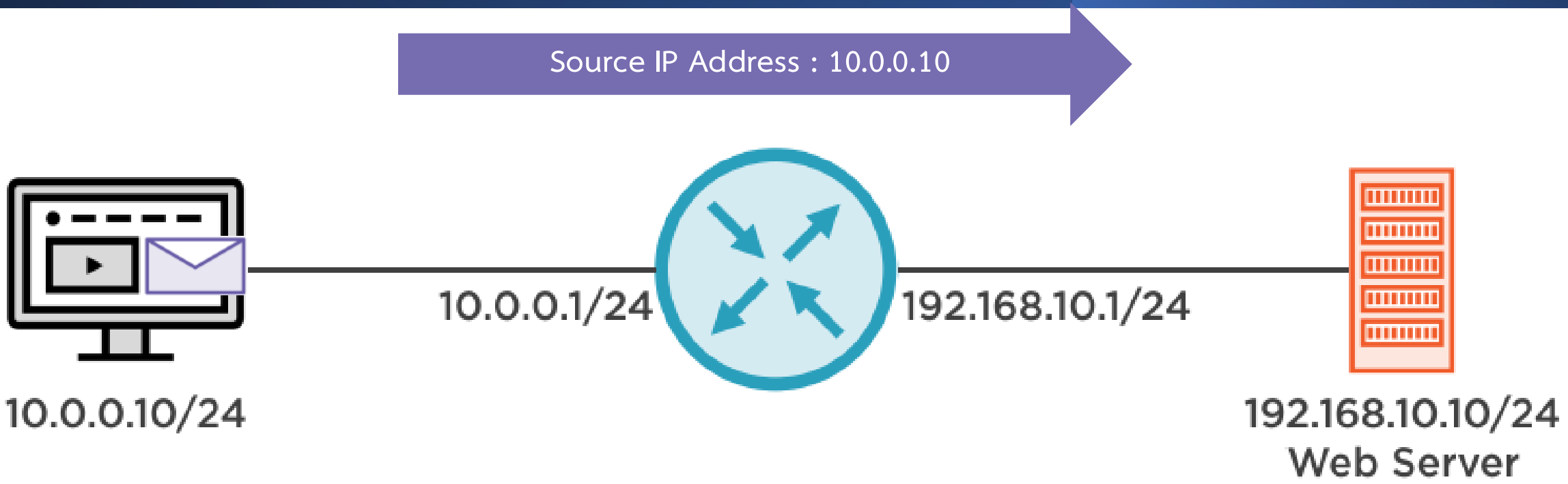
Standard Access Control Lists



1. สร้าง ACL

```
ip access-list standard TEST
permit host 10.0.0.10
```

Standard Access Control Lists

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permit host 10.0.0.10
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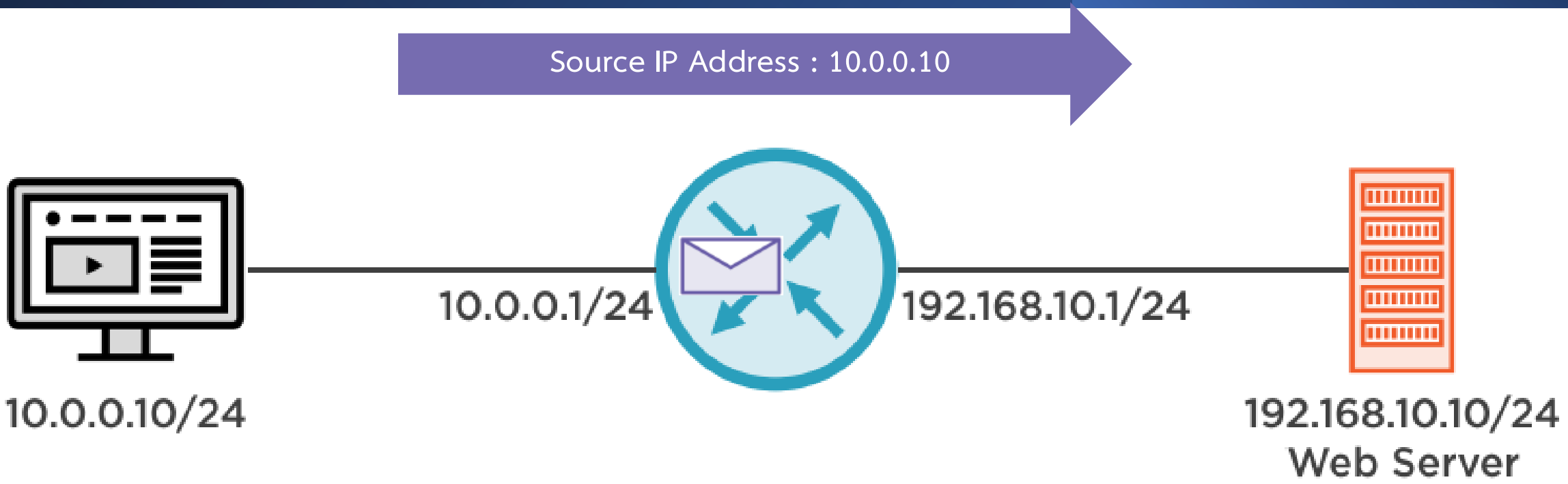
Standard Access Control Lists



ip access-list standard TEST

permit host 10.0.0.10

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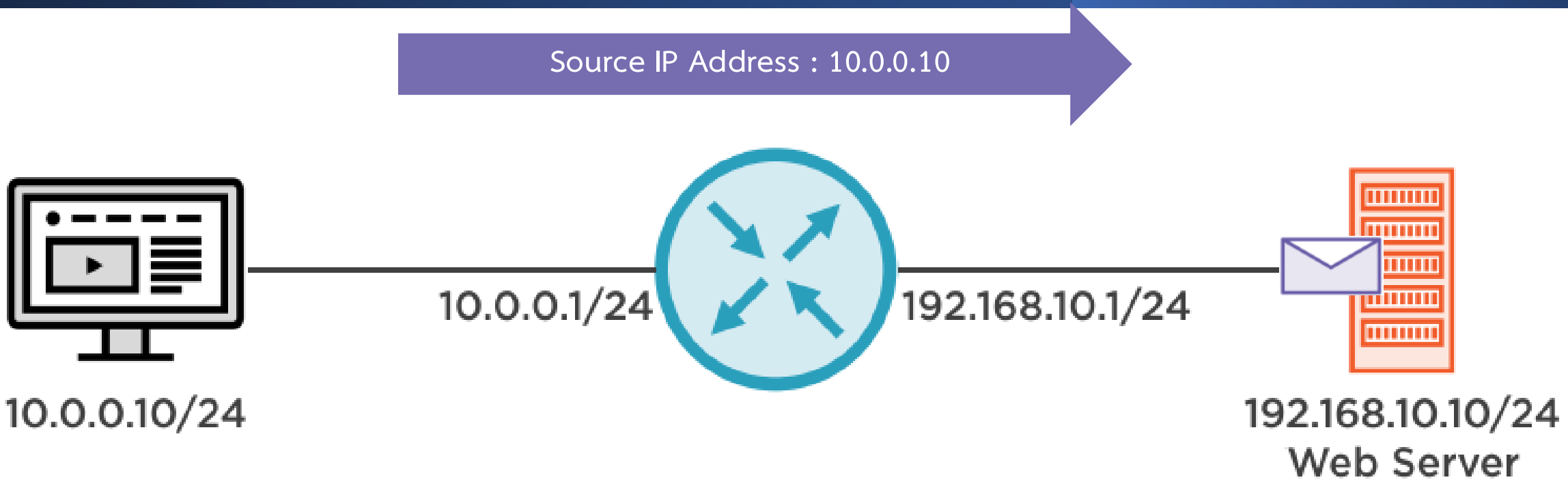


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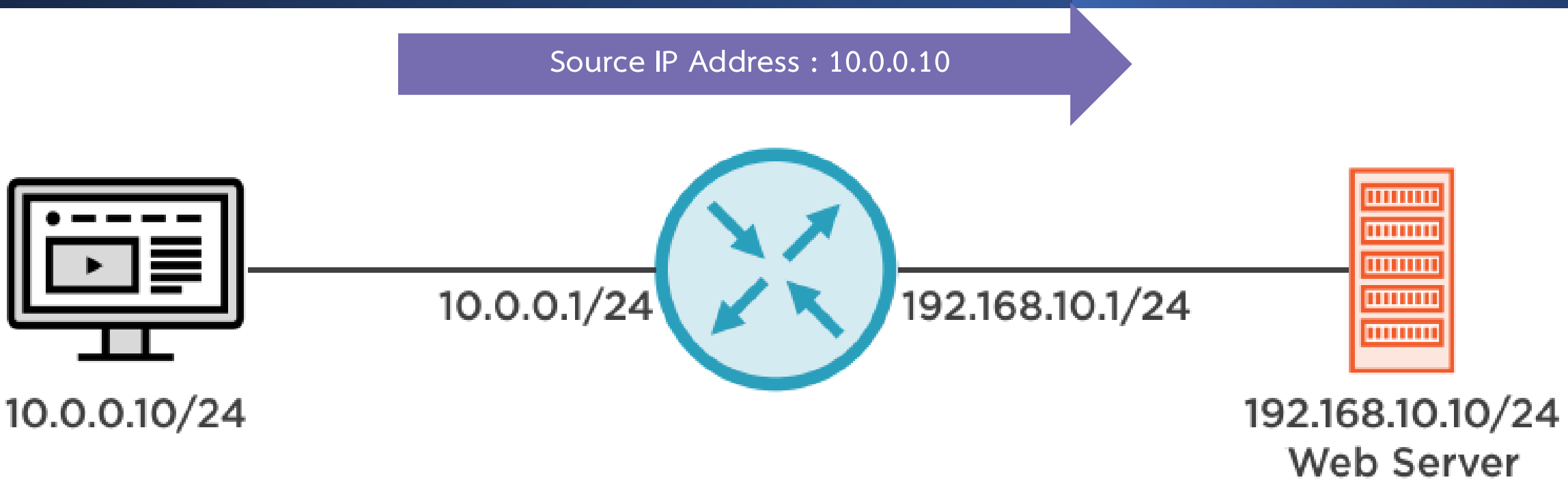


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permit host 10.0.0.10
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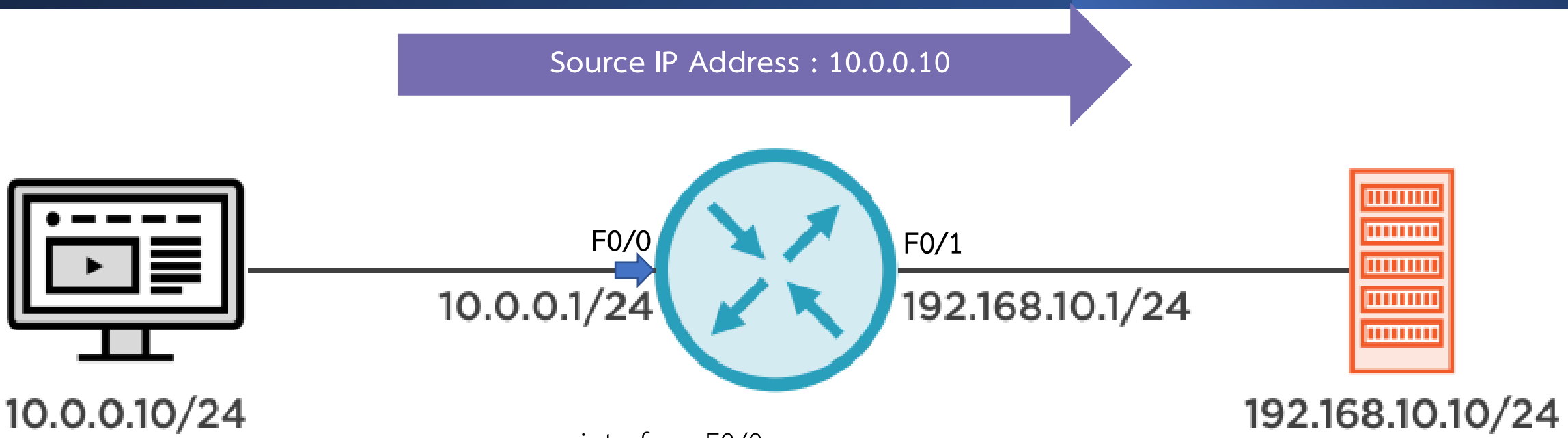
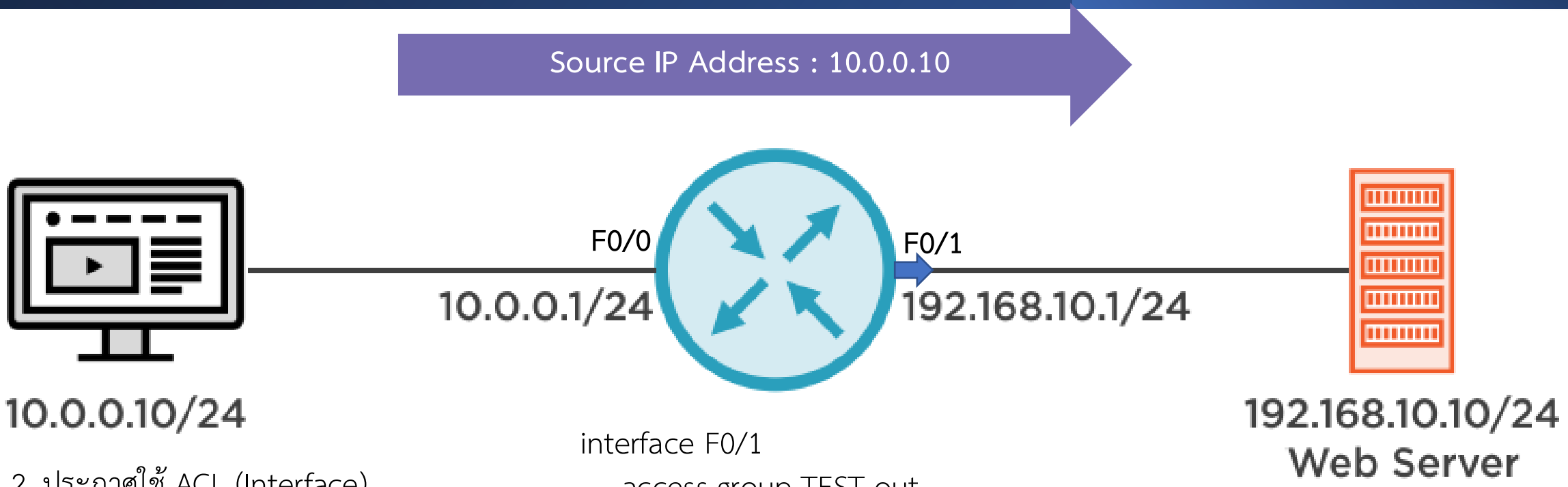


2. ประกาศใช้ ACL (Interface)

```
interface F0/0
    access-group TEST in
ip access-list standard TEST
    permit host 10.0.0.10
```

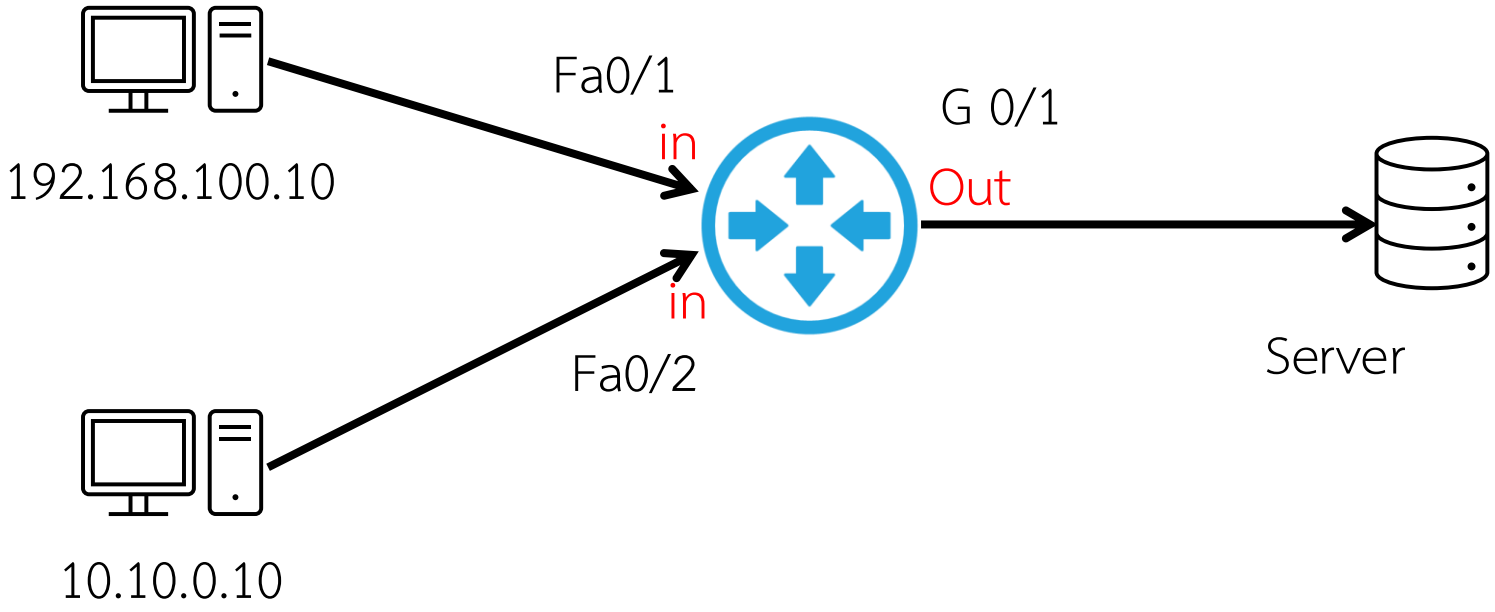
Standard Access Control Lists



2. ประกาศใช้ ACL (Interface)

```
interface F0/1
    access-group TEST out
ip access-list standard TEST
    permit host 10.0.0.10
```

การเลือกจุดใช้ ACL



Wildcard Mask

Subnet Mark

203.0.113.0/24

203.0.113.0 – 203.0.113.255

N	11001011	00000000	01110001	00000000
B	11001011	00000000	01110001	11111111
Subnet Mask	11111111	11111111	11111111	00000000

Wildcard Mark

203.0.113.0/24

203.0.113.0 – 203.0.113.255

N	11001011	00000000	01110001	00000000
B	11001011	00000000	01110001	11111111
Subnet Mask	11111111	11111111	11111111	00000000

The Wildcard Mask

00000000 00000000 00000000 11111111

0.0.0.255

Wildcard Mark

10.0.1.88 – 10.0.1.95

First	00001010 00000000 00000001 01011000
Last	00001010 00000000 00000001 01011111

Wildcard Mark

10.0.1.88 – 10.0.1.95

First	00001010 00000000 00000001 01011000
Last	00001010 00000000 00000001 01011111

Wildcard Mark

10.0.1.88 – 10.0.1.95

First	00001010	00000000	00000001	01011000
Last	00001010	00000000	00000001	01011111
Wildcard Mask	00000000	00000000	00000000	00000111

10.0.1.88 0.0.0.7

Wildcard Mark

10.0.0.64 – 10.0.0.79

First	00001010 00000000 00000000 01000000
Last	00001010 00000000 00000000 01001111

Wildcard Mark

10.0.0.64 – 10.0.0.79

First	00001010 00000000 00000000 01000000
Last	00001010 00000000 00000000 01001111

Wildcard Mark

10.0.0.64 – 10.0.0.79

First	00001010	00000000	00000000	01000000
Last	00001010	00000000	00000000	01001111
Wildcard Mask	00000000	00000000	00000000	00001111

10.0.0.64 0.0.0.15

Wildcard Mark

10.0.0.14 – 10.0.0.99

First	00001010	00000000	00000000	00001110
Last	00001010	00000000	00000000	01100011

Wildcard Mark

10.0.0.14 – 10.0.0.99

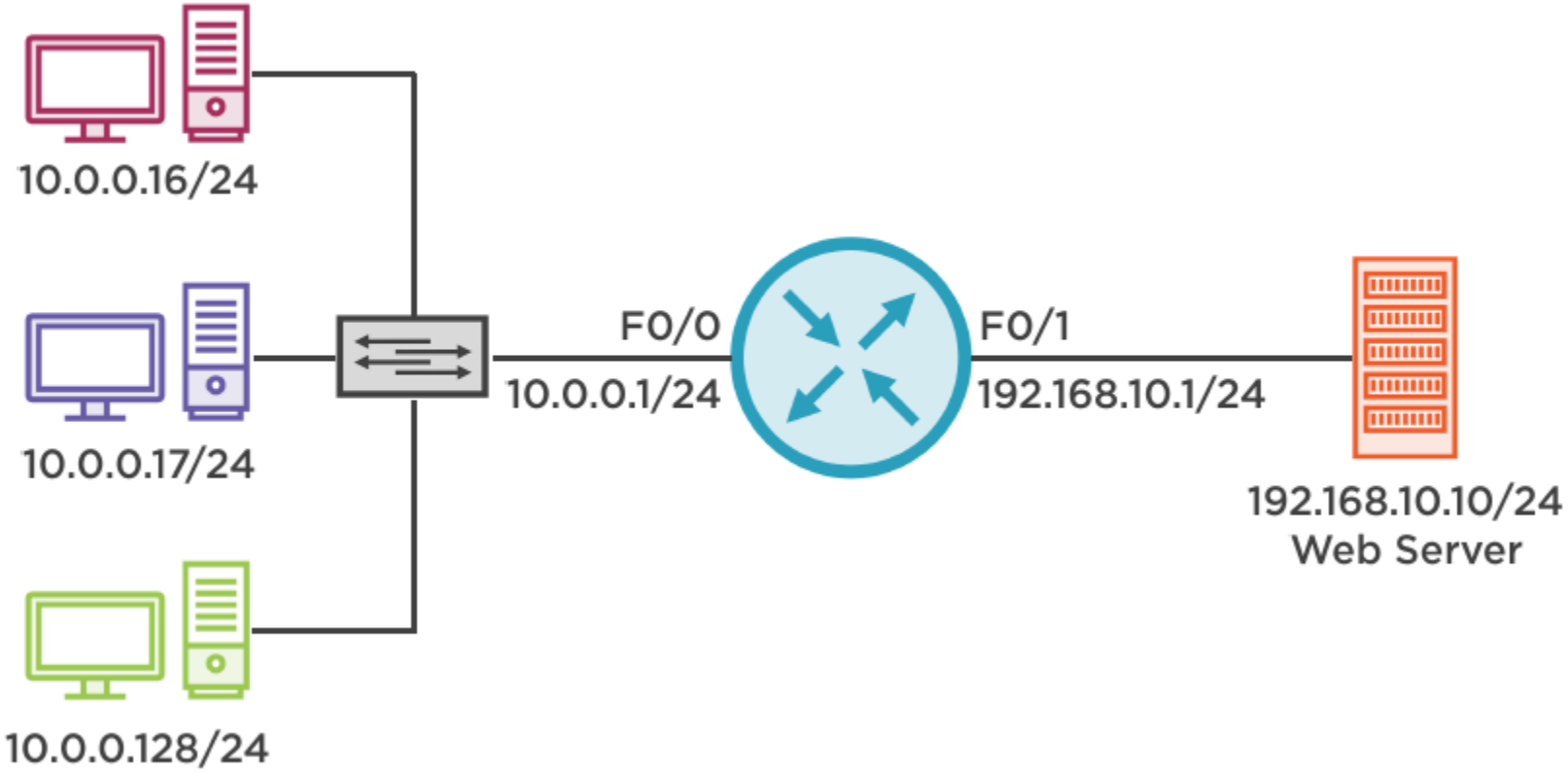
First	00001010 00000000 00000000 00001110
Last	00001010 00000000 00000000 01100011

Wildcard Mark

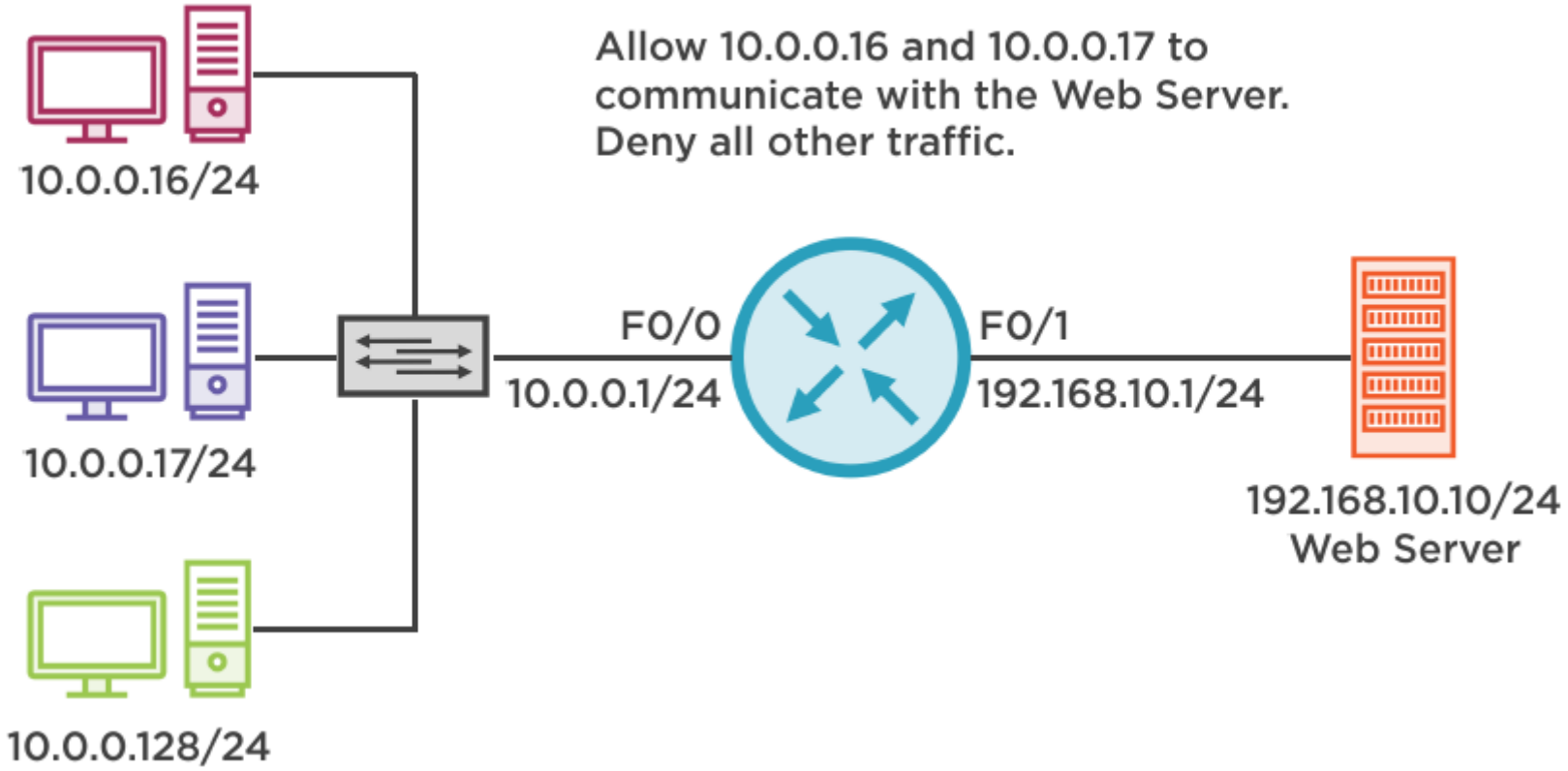
10.0.0.14 – 10.0.0.99

First	00001010	00000000	00000000	00001110
Last	00001010	00000000	00000000	01100011
Wildcard Mask	00000000	00000000	00000000	01111111

Standard Access Control Lists



Standard Access Control Lists



Standard ACL Syntax - Numbered

```
Router(config)#access-list <1-99> [permit/deny]<source IP> <w.c. mask>
                          <1300-1999>
```

OR

```
Router(config)#access-list <1-99> [permit/deny] host <source IP>
                          <1300-1999>
```

Standard ACL Syntax - Named

```
Router(config)#ip access-list standard <name>

Router(config-std-nacl)#[permit/deny]<source IP> <w.c. mask>
```

OR

```
Router(config-std-nacl)#[permit/deny] host <source IP>
```

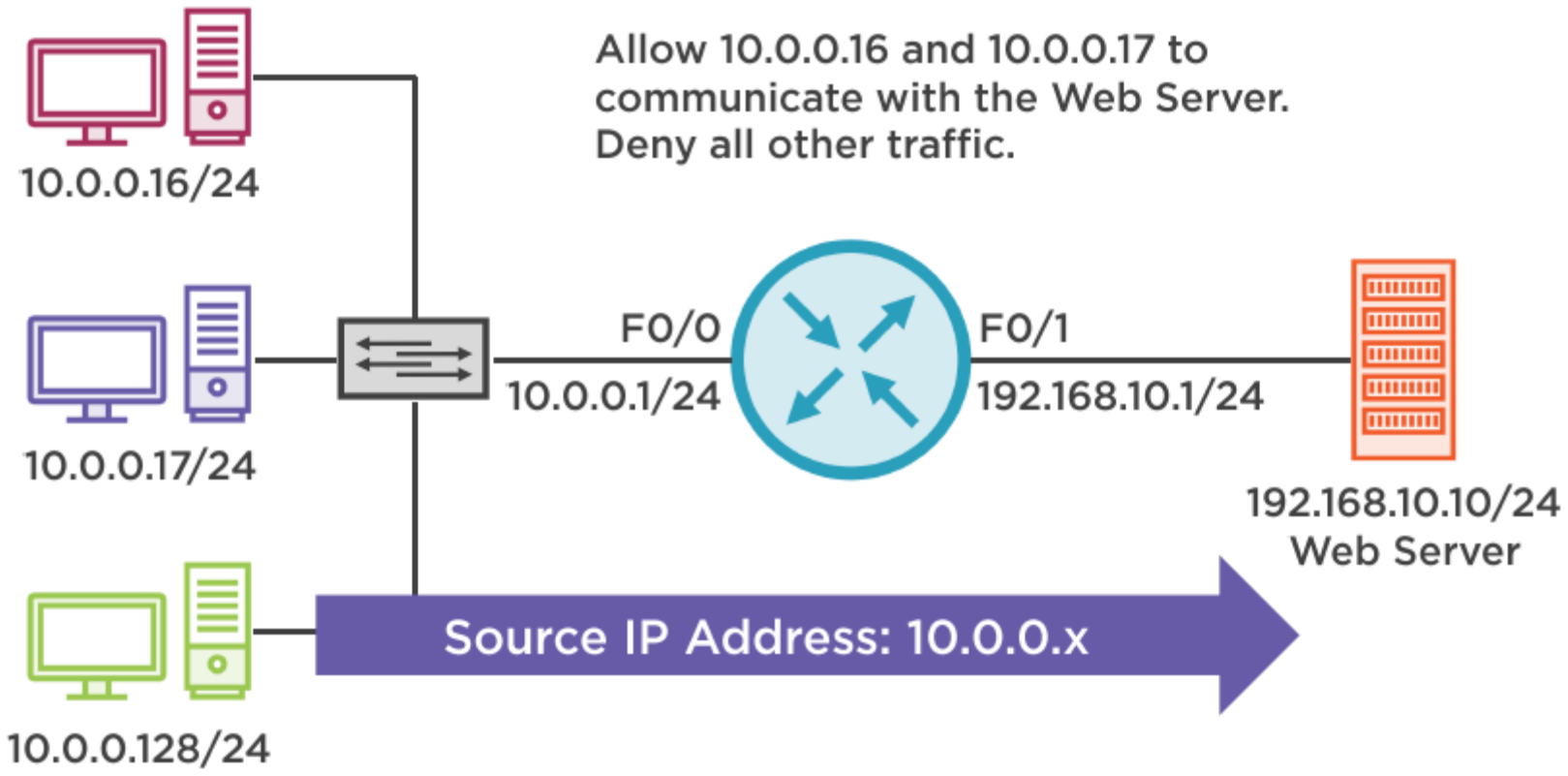
host <source IP> = <source IP> 0.0.0.0

Standard ACL Syntax - Named

Allow 10.0.0.16 and 10.0.0.17 to communicate with the Web Server. Deny all other traffic.

```
Router(config)#ip access-list standard Example1
Router(config-std-nacl)#permit 10.0.0.16 0.0.0.1
Router(config-std-nacl)#deny any
```

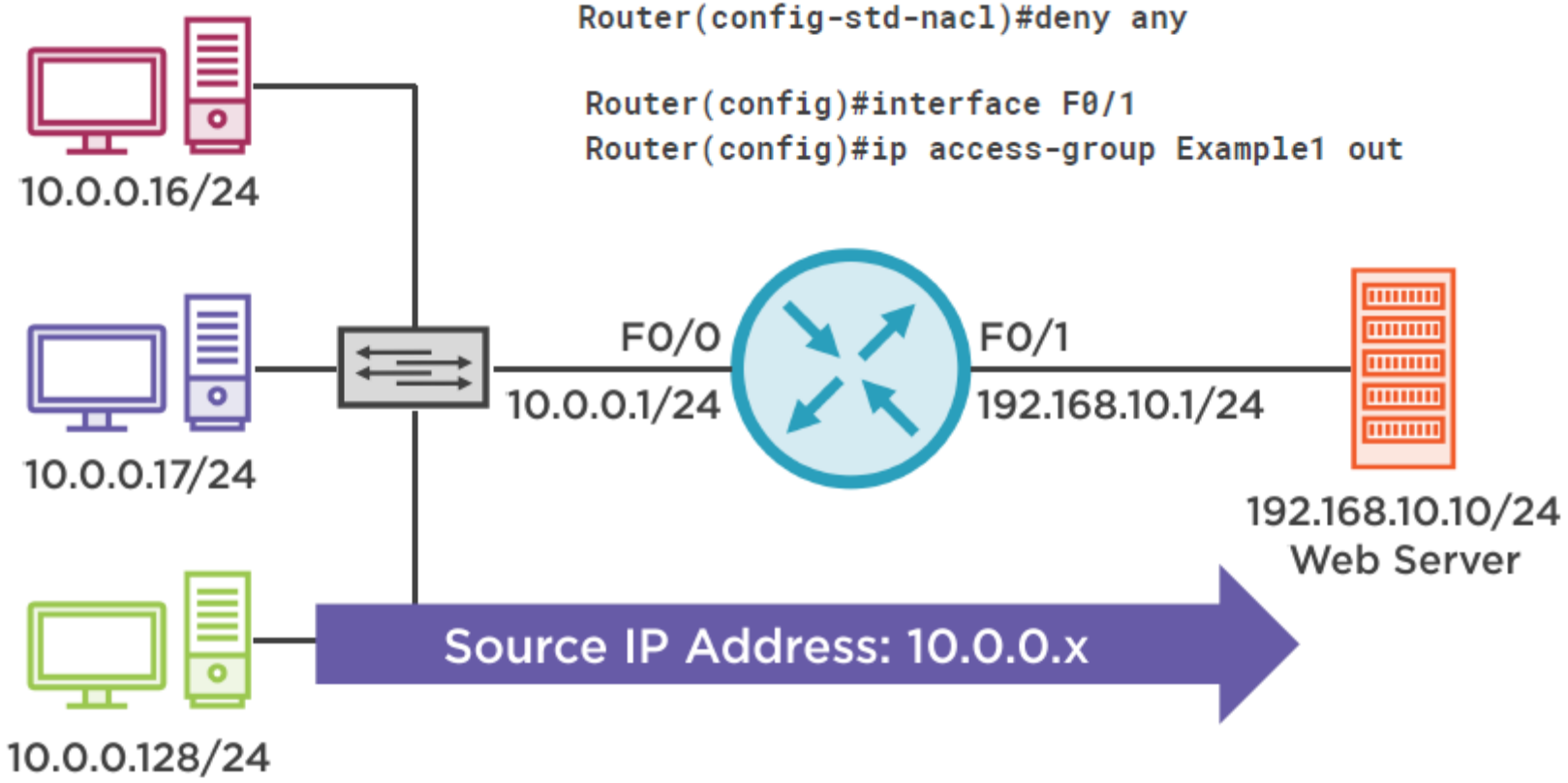
Standard Access Control Lists



Standard Access Control Lists

```
Router(config)#ip access-list standard Example1
Router(config-std-nacl)#permit 10.0.0.16 0.0.0.1
Router(config-std-nacl)#deny any

Router(config)#interface F0/1
Router(config)#ip access-group Example1 out
```



Extended Access Control List

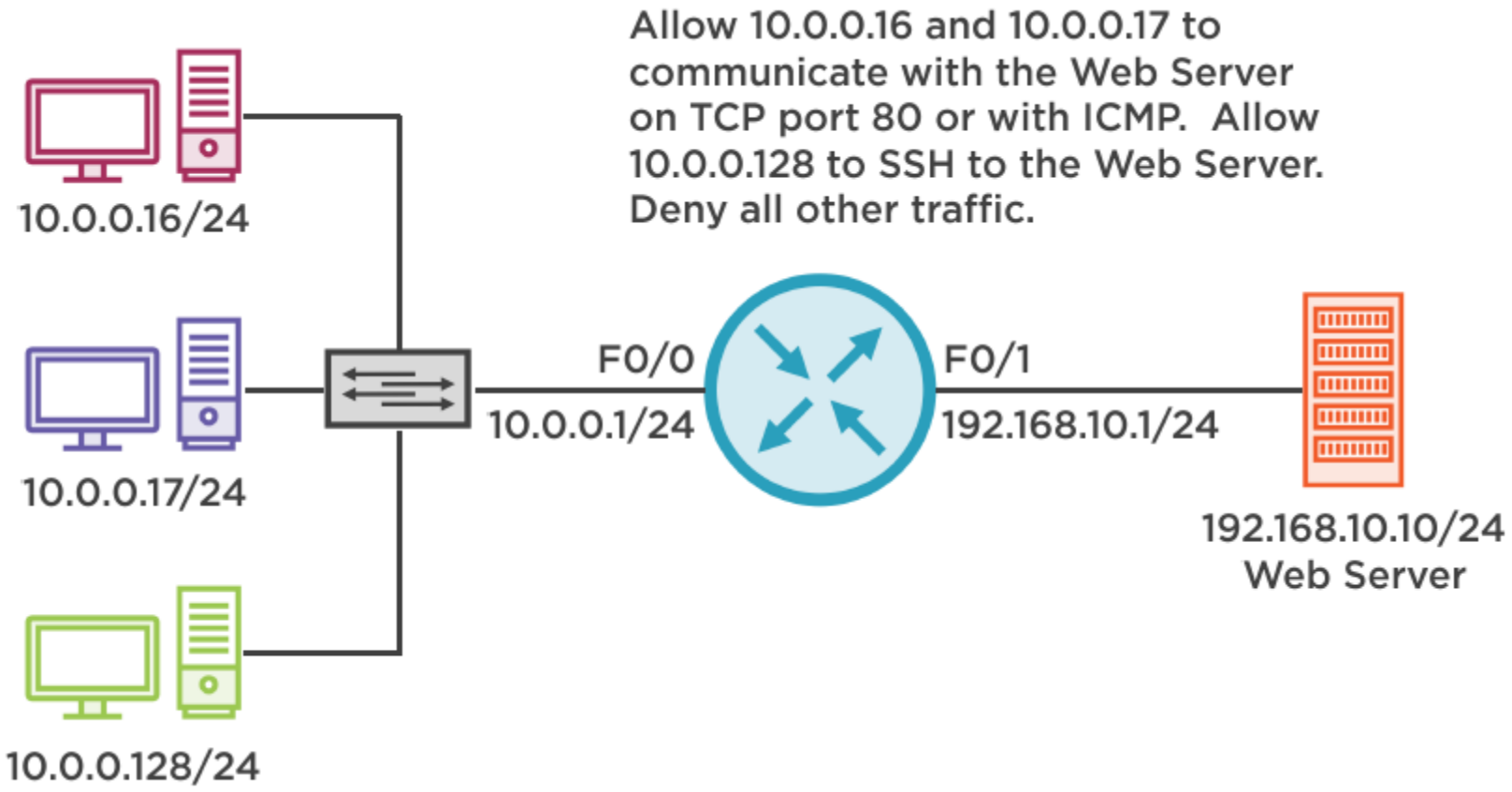
Extended ACL Syntax

```
access-list extended <100-199, 2000-2699> [permit/deny] [ip/tcp/udp/icmp]
<source IP> <w.c. mask> {eq <source port number>}
<destination IP> <w.c. mask> {eq <destination port number>}
```

```
ip access-list extended <name>

[permit/deny] [ip/tcp/udp/icmp] <source IP> <w.c. mask>
{eq <source port number>} <destination IP> <w.c. mask>
{eq <destination port number>}
```

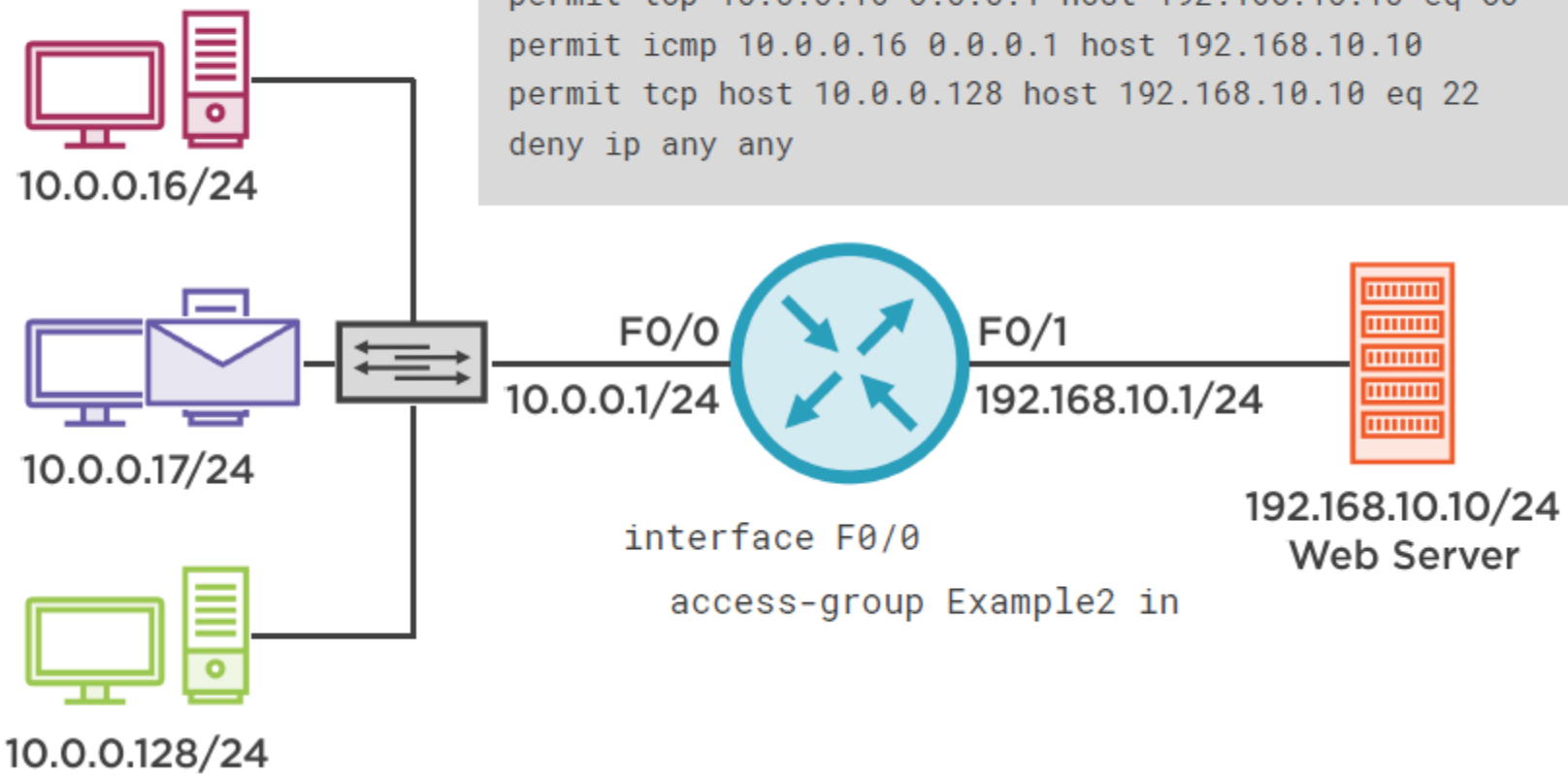
Extended Access Control List



Extended Access Control List

```
ip access-list extended Example2

permit tcp 10.0.0.16 0.0.0.1 host 192.168.10.10 eq 80
permit icmp 10.0.0.16 0.0.0.1 host 192.168.10.10
permit tcp host 10.0.0.128 host 192.168.10.10 eq 22
deny ip any any
```



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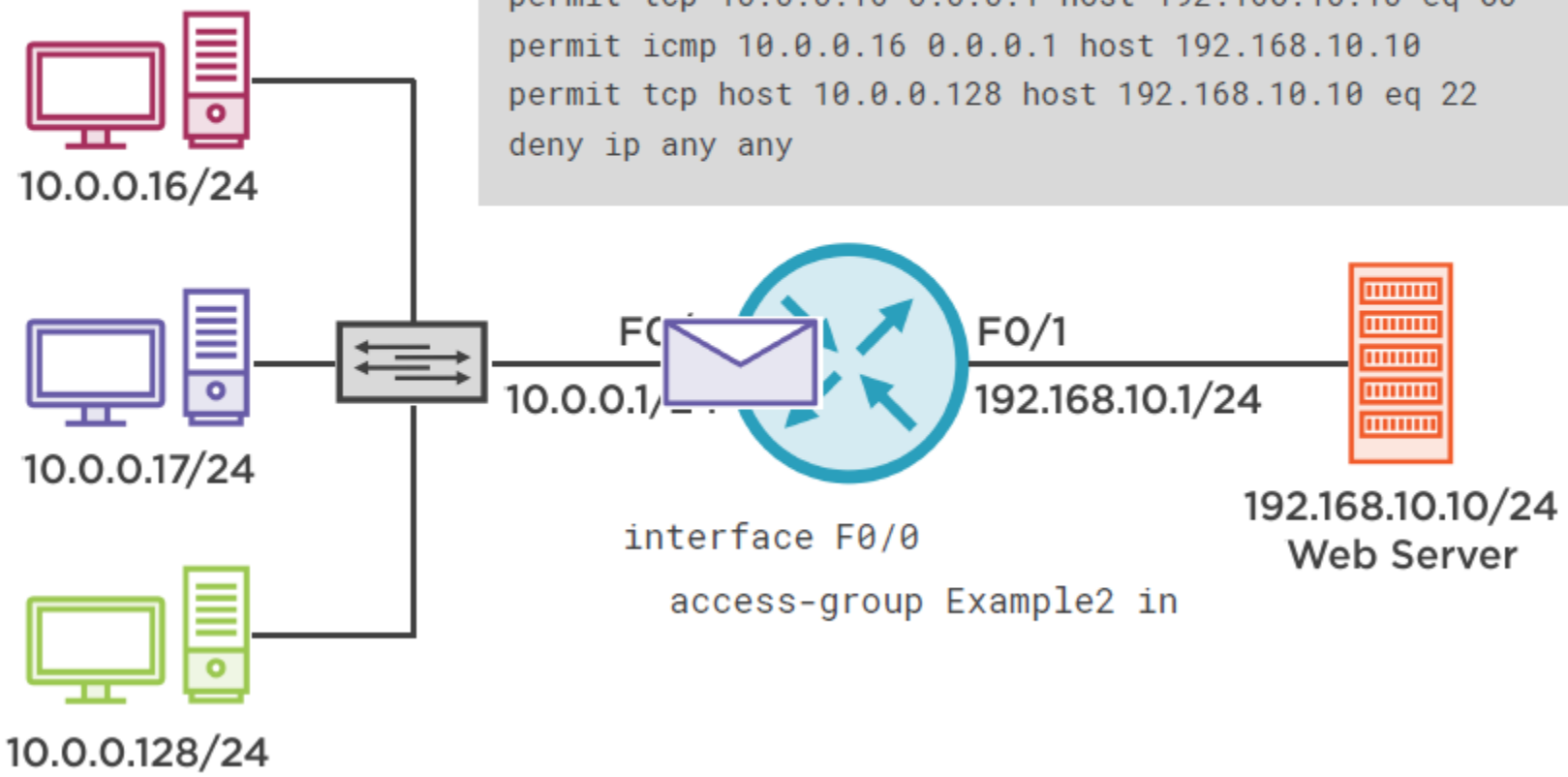

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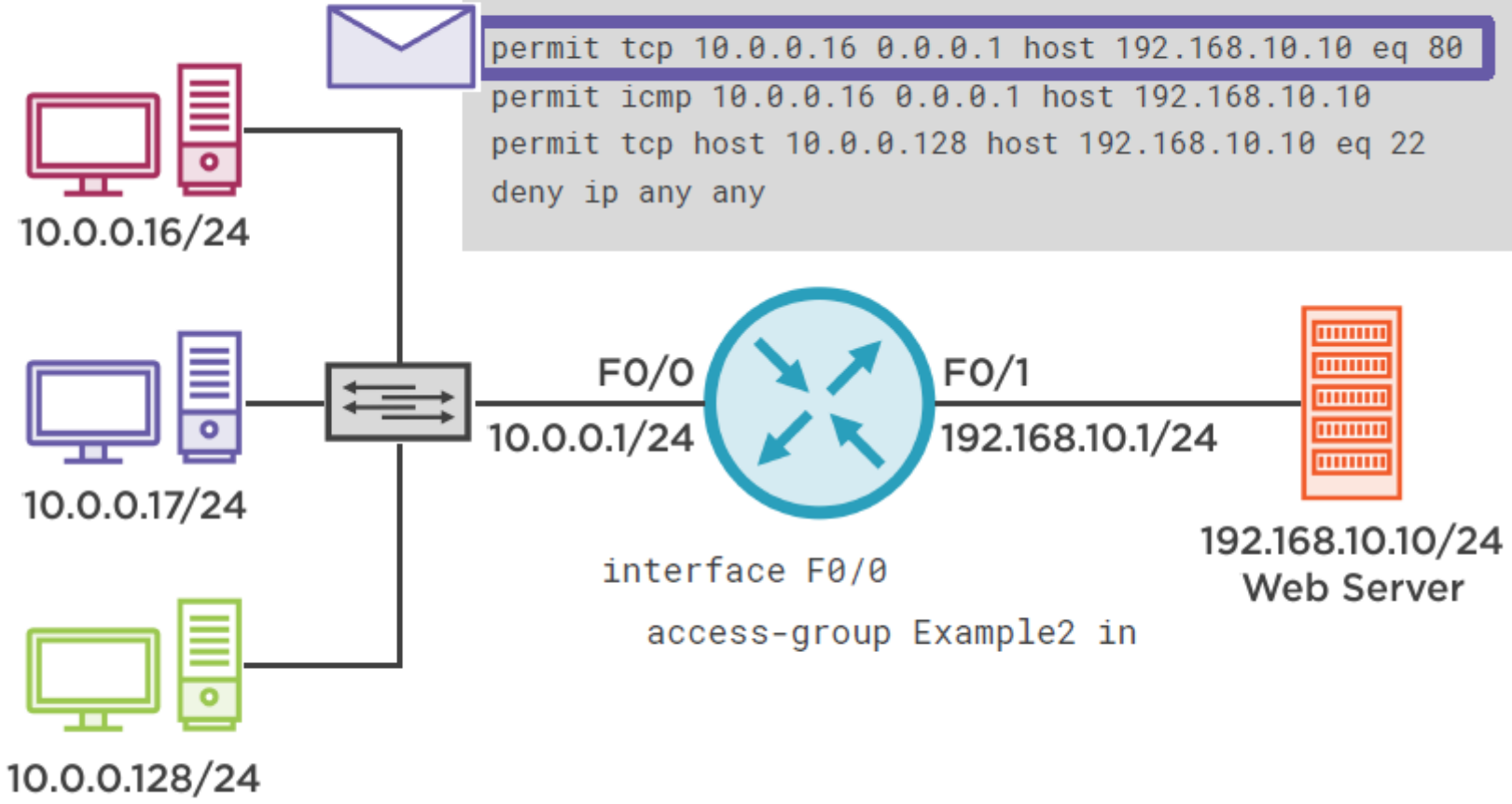


```
interface F0/0
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```

Extended Access Control List

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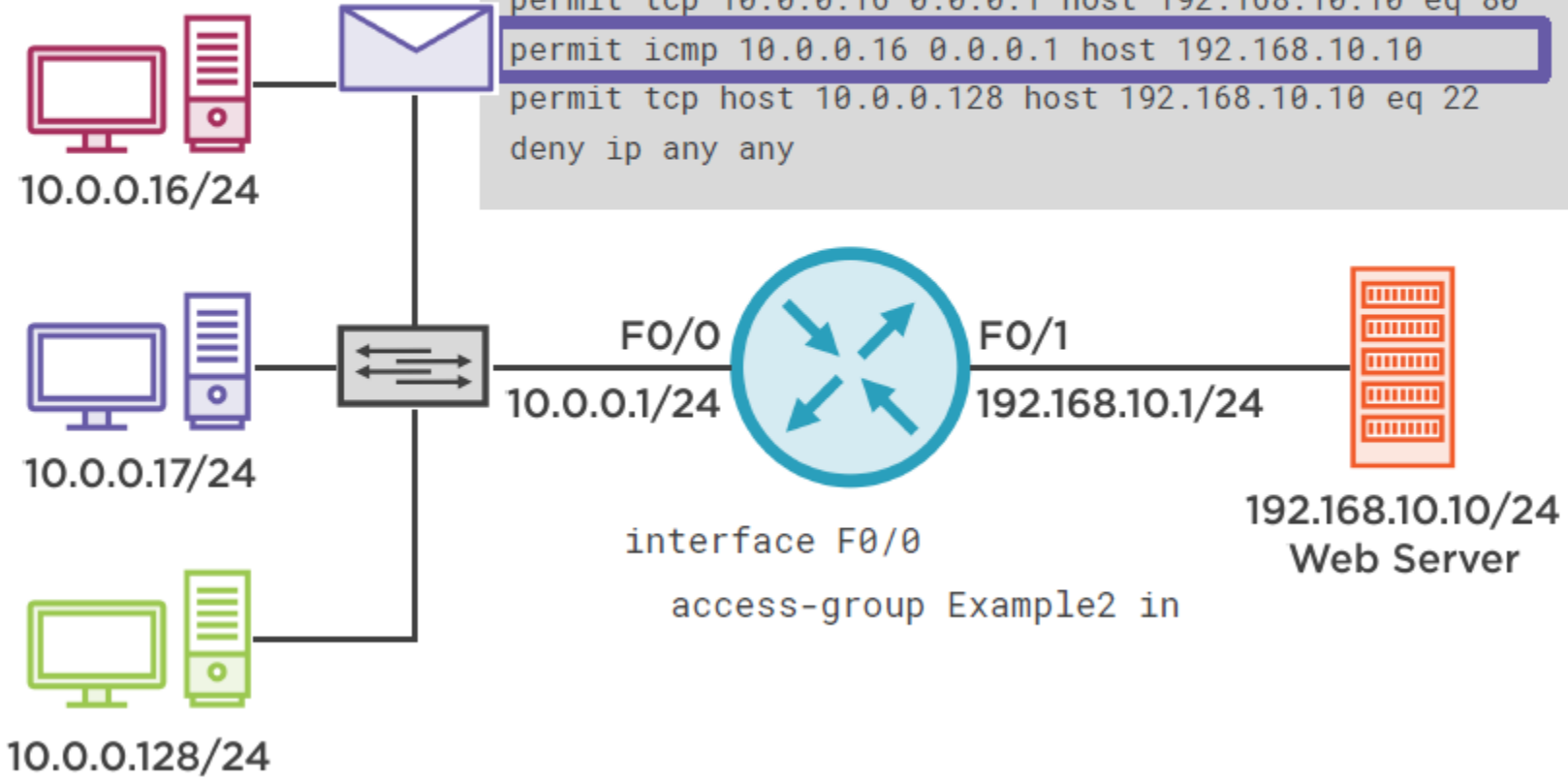


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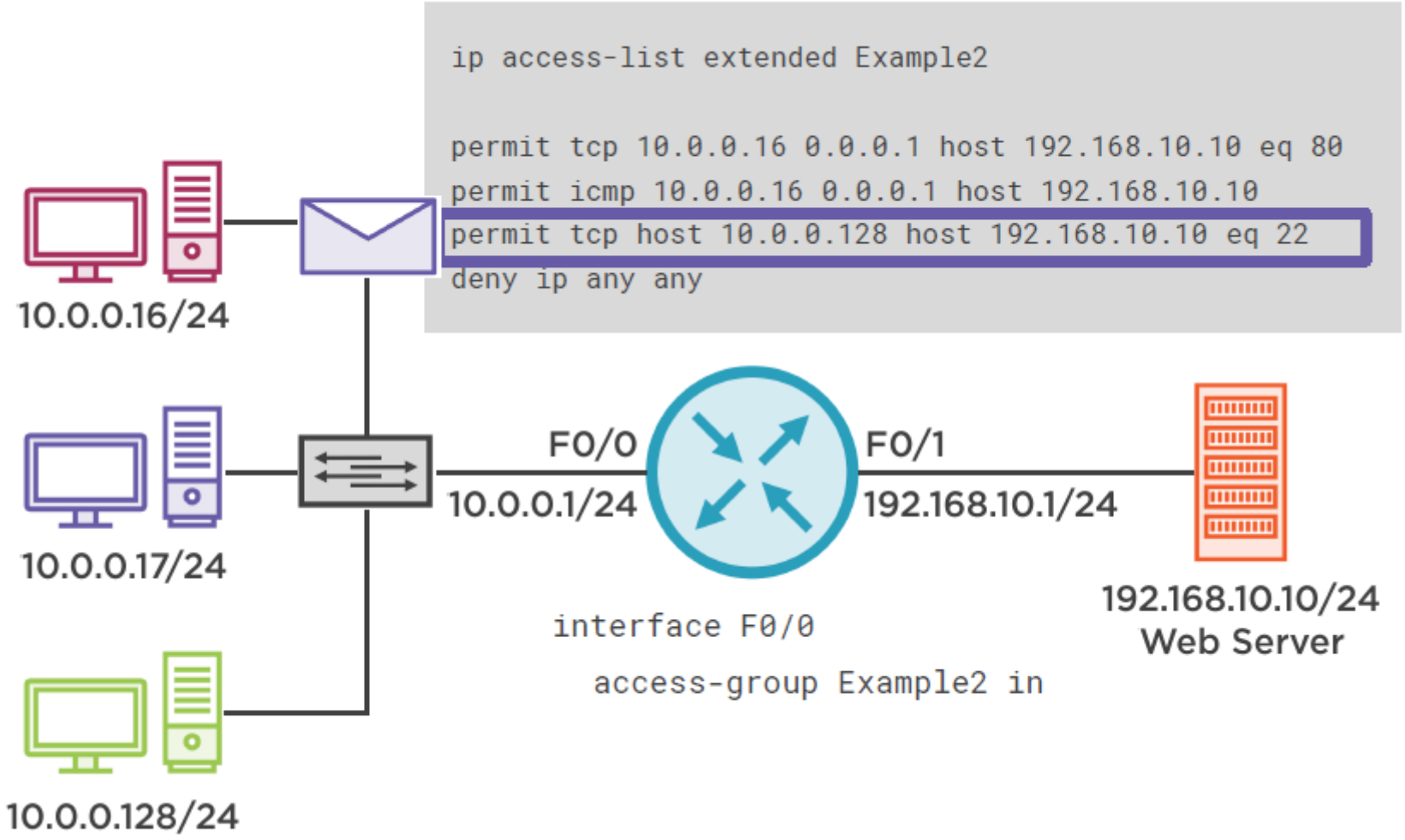
Extended Access Control List



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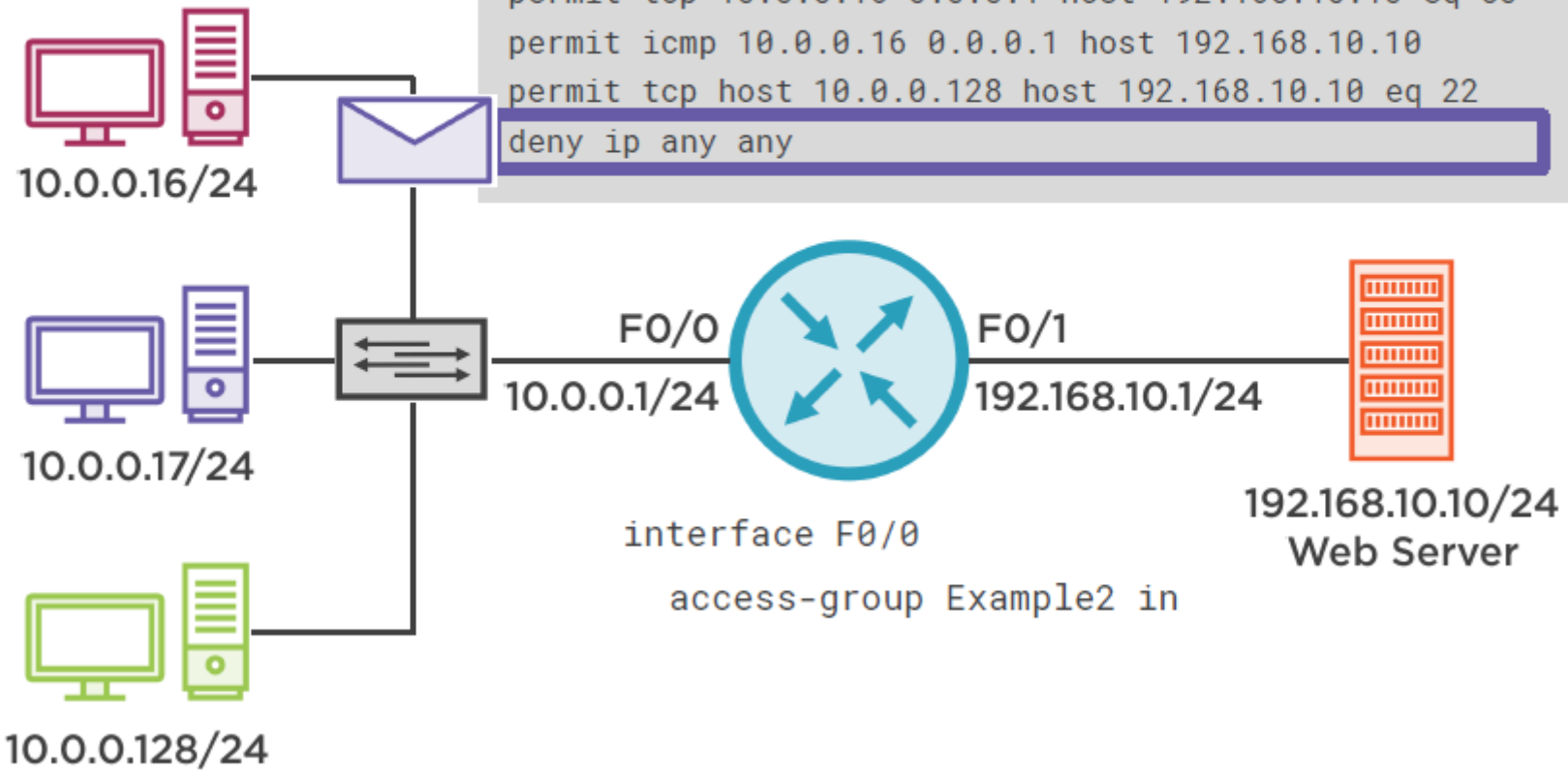


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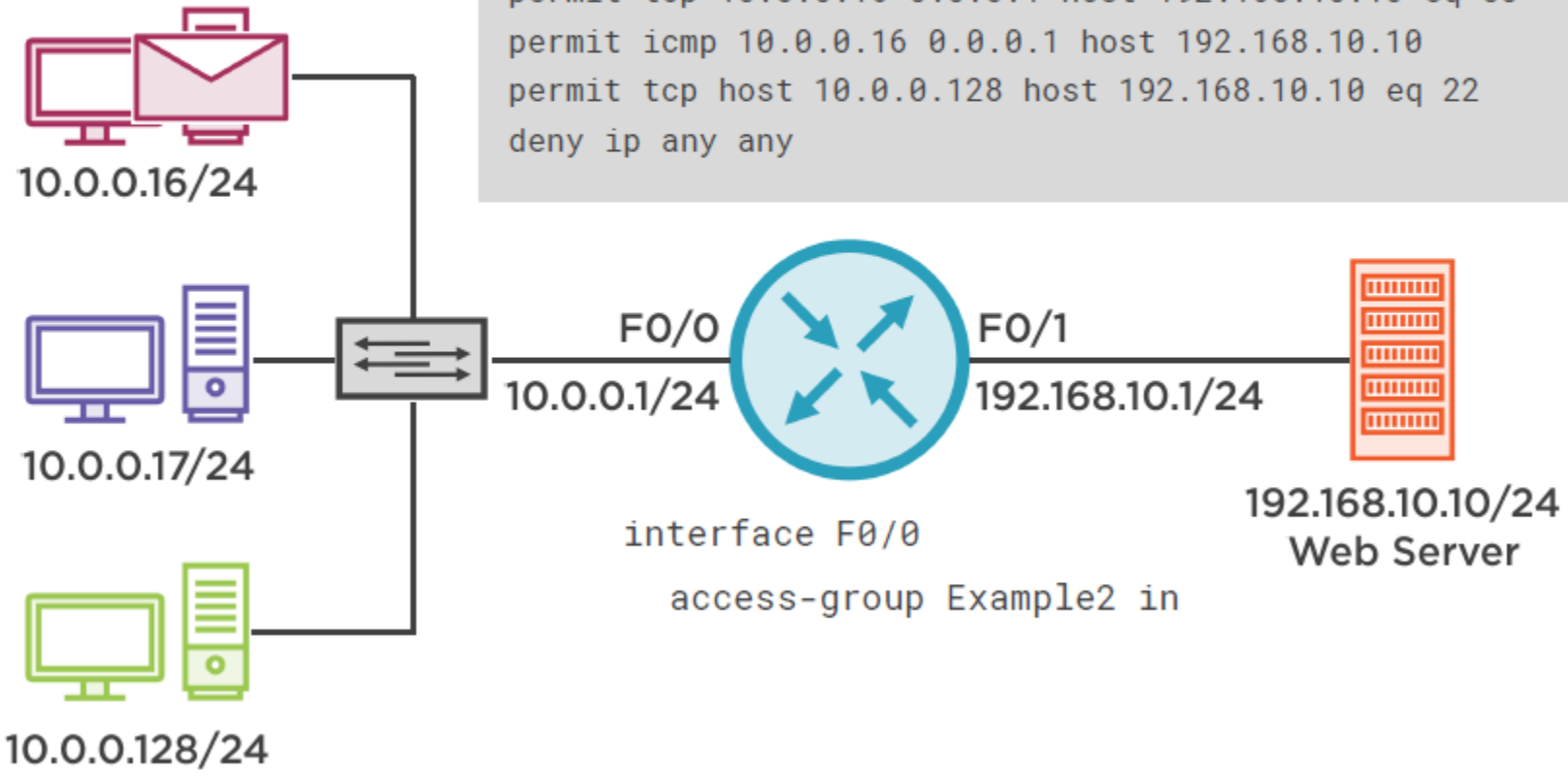

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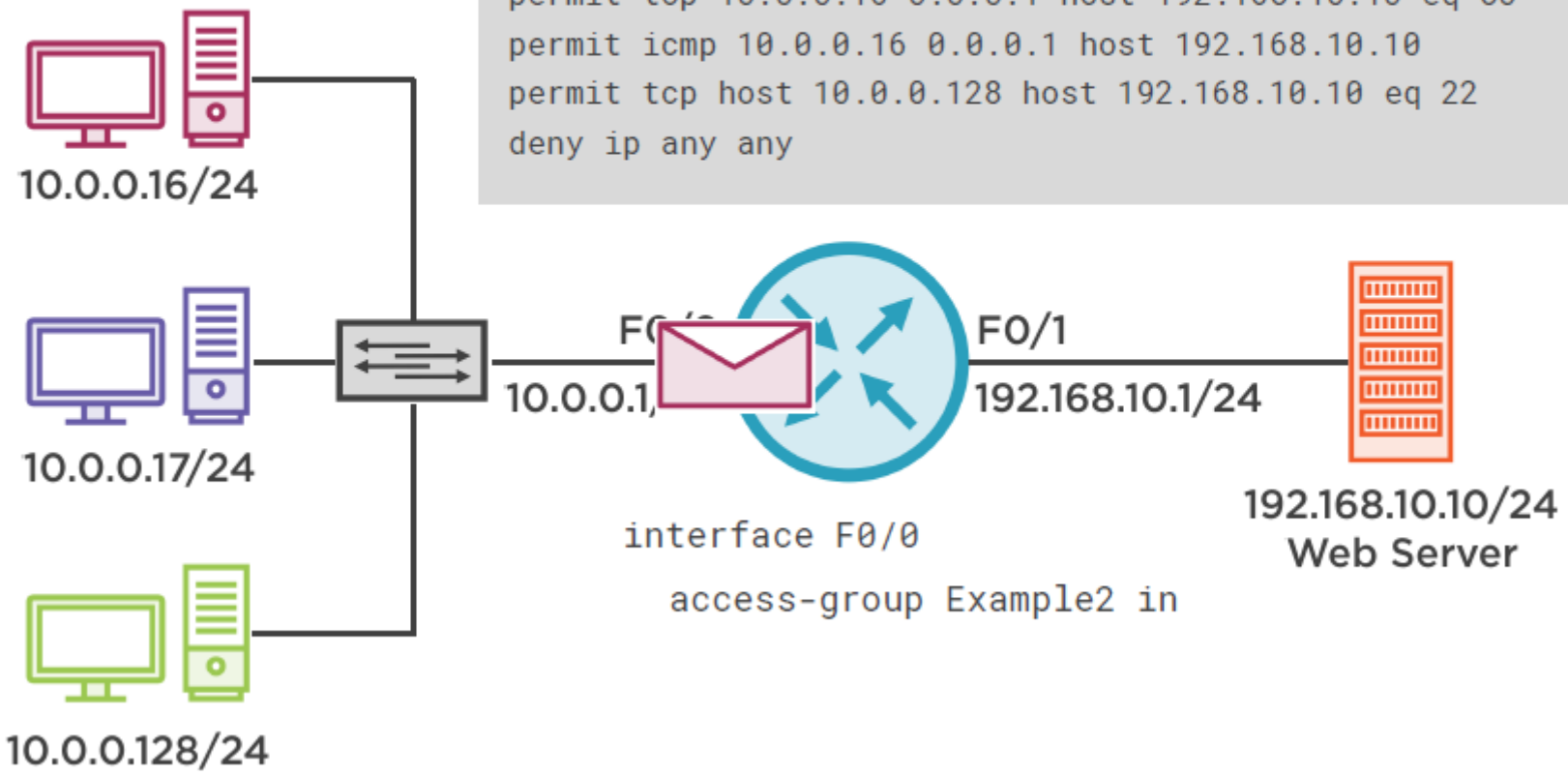


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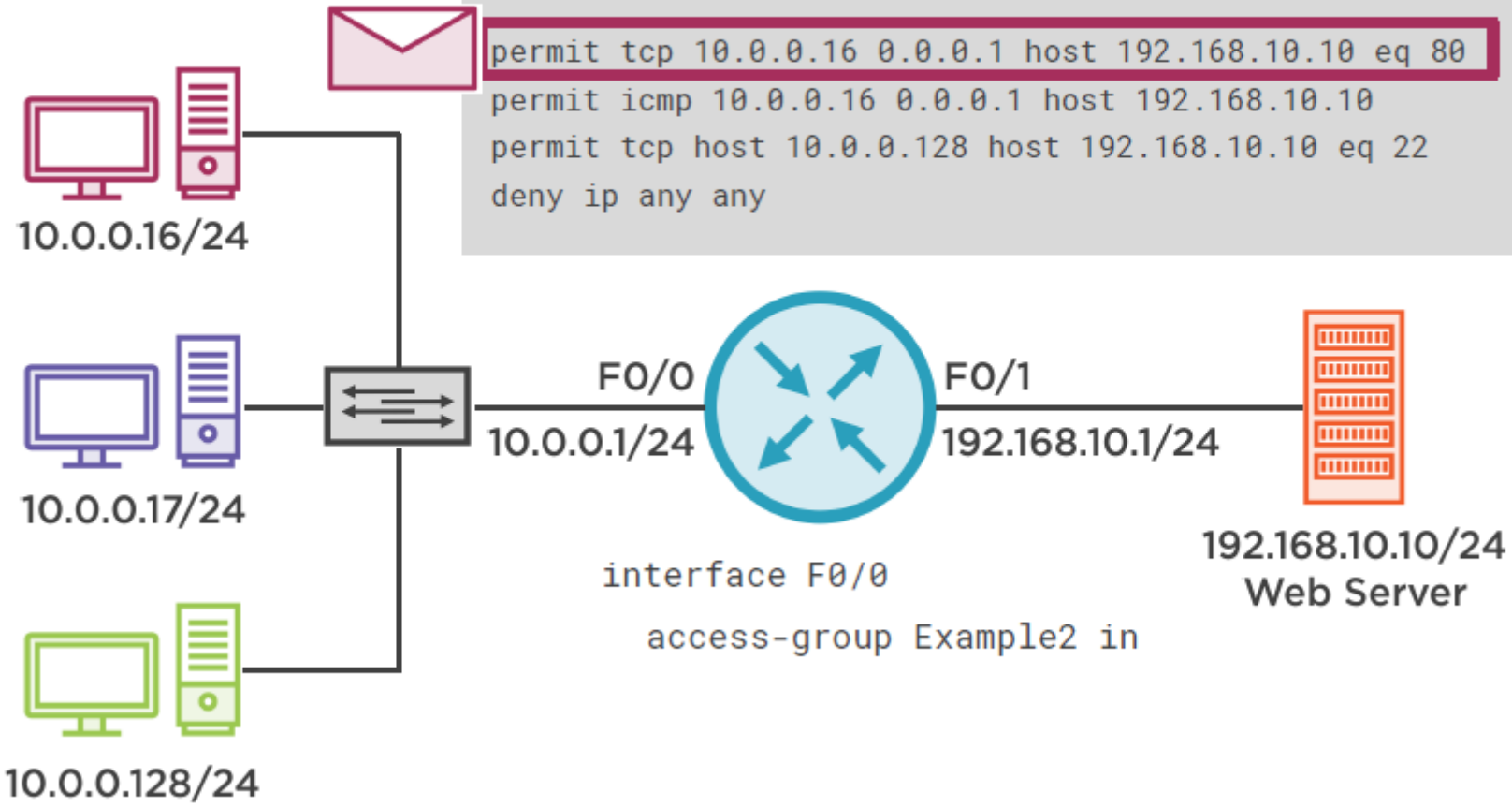


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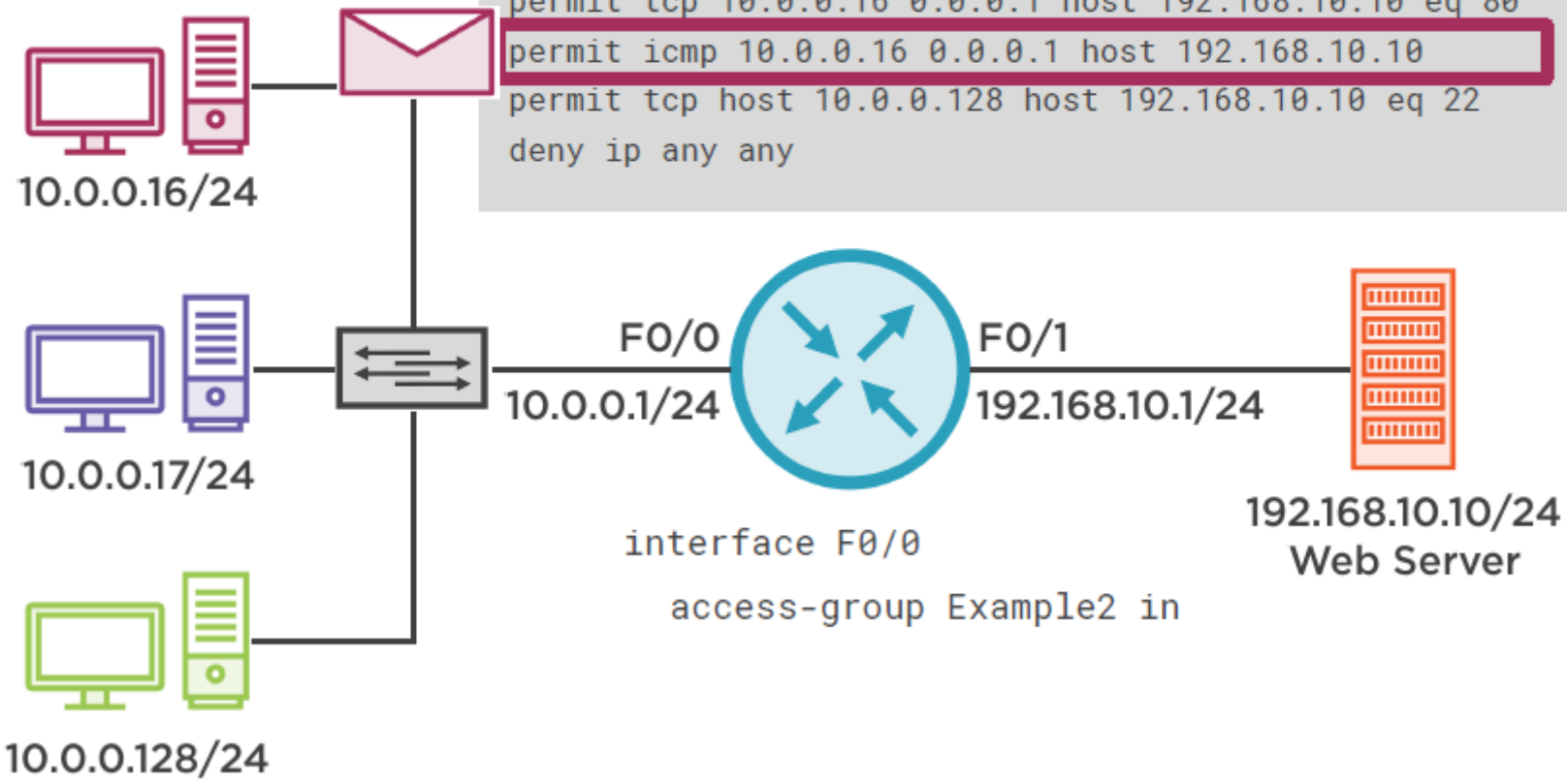


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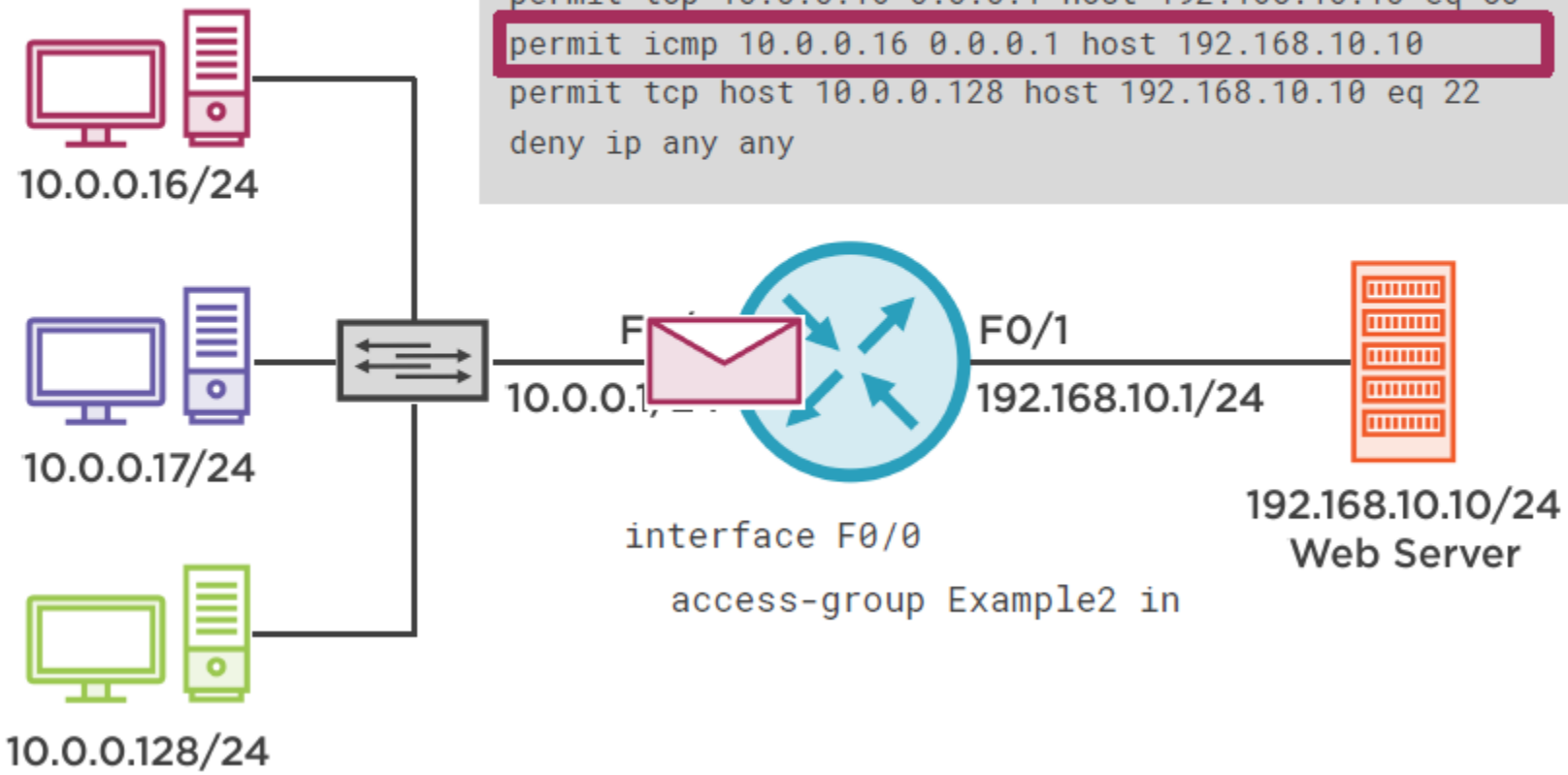


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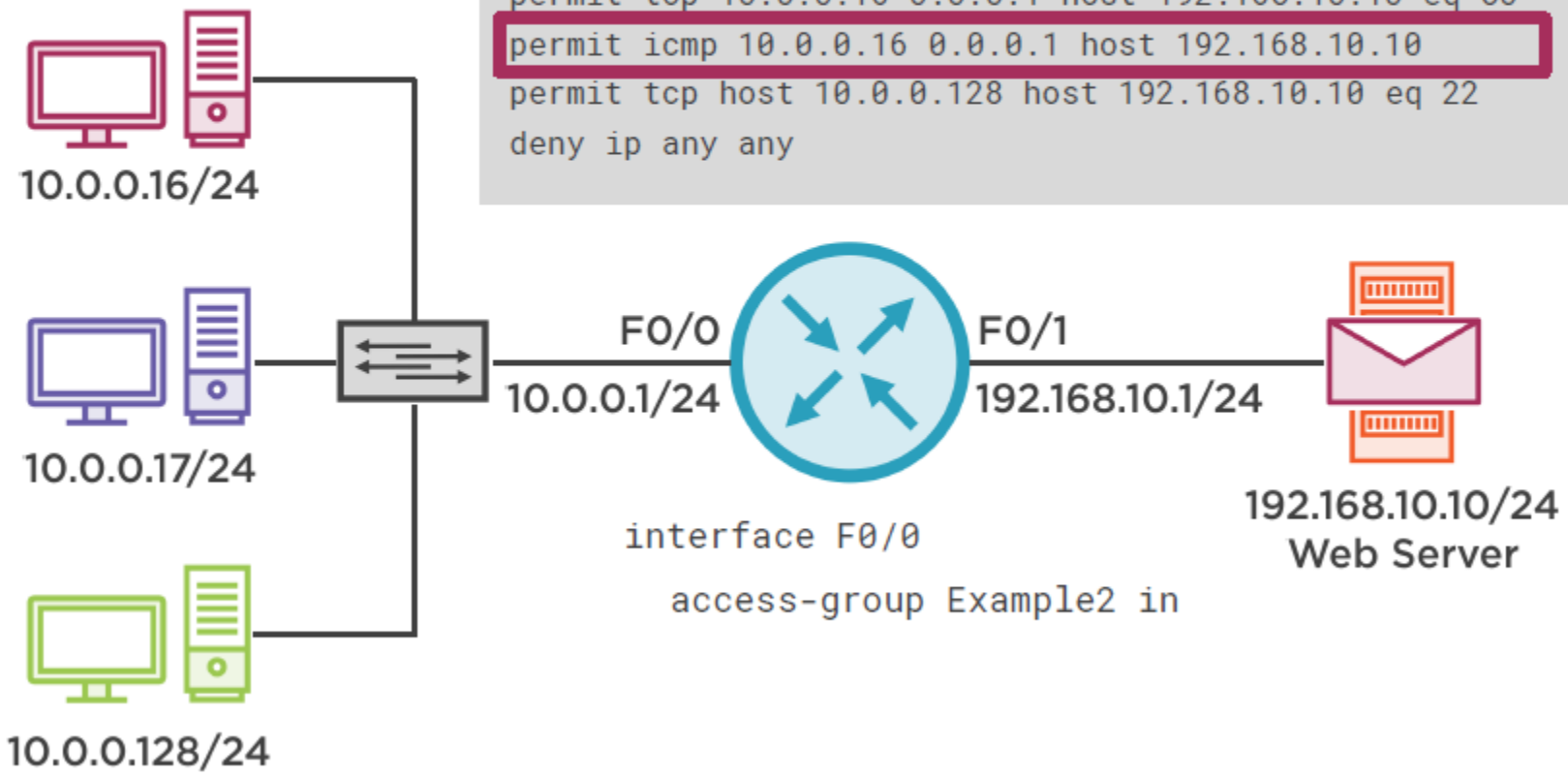


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Simulation